Air Circulation Robot

By Andy Liu A17112518 anl046@ucsd.edu, Jeffrey Lee A17442839 jsl019@ucsd.edu

We plan on building a device that circulates, redistributes, and filters air within its proximity through a fan that also serves as an air purifier. This device will act as an autonomous robot that seeks out areas of the indoor environment that have higher temperatures, humidity, and higher concentrations of air particles. This is useful because certain parts of the room can be hot, stuffy, and generally difficult to breathe in due to poor air conditions (high temperatures, humidity, concentrations of air particles, etc.). Furthermore, the demand for air conditioning and purifying systems in indoor environments have been in increasing demand due to circumstances from climate change, inherent or developed sensitivities to airborne particles (whether from allergies or respiratory conditions), and in response to the rise in respiratory infectious diseases (notably COVID-19 and other respiratory viral illnesses) that can spread through water particles in the air; however, these systems can be costly, consume large amounts of energy, and may not evenly distribute across the environment to effectively improve air conditions throughout.

This project will prove to be challenging as we will require to have many components interfacing with each other: sensors receiving data, actuators to respond to the measurements, and software to monitor the sensor input and adjust how the device behaves over time. In addition to responding to its physical environment (to ensure that the device does not run into objects/walls and can navigate the local environment), the device must also decide on where to go based on other input data (based on temperature, humidity, and air quality). An algorithm will need to be developed for the system and implemented so that it can securely and efficiently interface with both the motors and fans.

Hardware Components

The hardware components that we expect to implement and interface with are as follows, along with how we intend to or will obtain them:

- 1 Ultrasonic Sensor (37 in 1)
- 1 GY-521 Module (37 in 1)
- 1 Avoidance (37 in 1)
- 1 PM2.5 Sensor and 2.54 Adapter Kit (Checkout part from CSE 147 Lab Inventory)
- 1 Temperature & Humidity Sensor (37 in 1)
- 2 DC Motors (Robot Chassis Kit from Amazon)
- 1 Motor Driver (Amazon)
- 3 Wheels (Robot Chassis Kit from Amazon)
- 1 Chassis (Robot Chassis Kit from Amazon / 3D-print our own)
- 1 Carbon Filter Sheet (Amazon)
- 1 Fan (Amazon)

Software Components

We will create a mapping of the room's temperature over time and determine which parts are prone to increasing in temperature. We also want to map out where the air particles are most dense. We will start at the corner of the room, and map that point to (0,0). All other locations that we move to in the room will be an area +x and +y. At each of these locations, we will measure the data and map it to that location. We

want to see where the trends are, so we will map each of these values to a list. We want to go over the data that we mapped and make sure there are no outliers. In the end, we can go over the data and check where the temperature, humidity, and air particles are greatest in the room. As we go through the room, we will also be using the avoidance sensor to avoid running into obstacles.

Metrics

We will measure success by checking the heat and air particle distribution before and after the device sweeps the room. We want the median and average to be close to one another. We also want the maximum and minimum temperature to be within 1 degree fahrenheit of each other, and ideally reduce the range between the maximum and minimum concentration of air particles to be by at least 10%. If the air quality is observably more pleasant and desirable, then that would be the clearest indication of the success of the capabilities of our device.

Experiments & Expected Results

To experiment with our project, we will test our device under various conditions and environments with expectation that upon testing our device in those conditions, the measurements of temperature, humidity, and air quality improve:

- Test the devices basic navigation capabilities, ensuring that it can detect obstacles and maneuver accordingly
- Test the device's ability to respond to air quality conditions by scattering dust (or some other particulate matter) into the air and seeing if:
    1. The device eventually navigates to that area detecting high concentrations air particles
    2. Is able to successfully filter / distribute the particulate matter in that location
- Test the devices ability to respond to temperature and humidity conditions in a location where a steamer/container of boiled water/humidifier is placed and seeing if:
    1. The device eventually navigates to that area detecting high temperatures and humidity
    2. Is able to successfully improve temperature and humidity conditions in the location
- We will add any more tests that we deem necessary to evaluate our device's performance

Deliverables by Mid-Project Report

By the mid project report, the hardware should be working properly. We should be able to sense the temperature, heat, and air particles in each location. We should be able to move our device while properly avoiding collisions. The fan should be able to spin on command.

Final Demo & Report

In the final demo, we will show what a room's temperature, humidity and air particles are like before and after. We will show the device moving around the room, and the data it collects as it goes along the room. We will report if we were able to meet or exceed our metrics. If we fail to meet our metrics, we will explain what went wrong, and how we could improve it. We also want to report on our process, and where we think we could have improved.